

As the water flows over the moss-covered rock, it divides into myriad tiny rivulets to form a delicate fringe dropping into the underlying pool.

Optimisticheskaya

Location Ternopil Oblast, western Ukraine

With 140 miles (230 km) of passageways, this is the longest gypsum cave system in the

world—and the longest cave of any description in Eurasia. Despite this, it exists in layers of gypsum just 65 ft (20 m) thick. Gypsum is a soft sulfate mineral, which is more soluble than limestone and readily dissolves when in contact with slightly acidic rainwater to form underground features similar to those in limestone caves. Optimisticheskaya has an extraordinarily dense network on several levels, giving rise to its name of “the maze cave.” It is accessible only through a single entrance, dug through the floor of a doline in 1966 by local cavers.

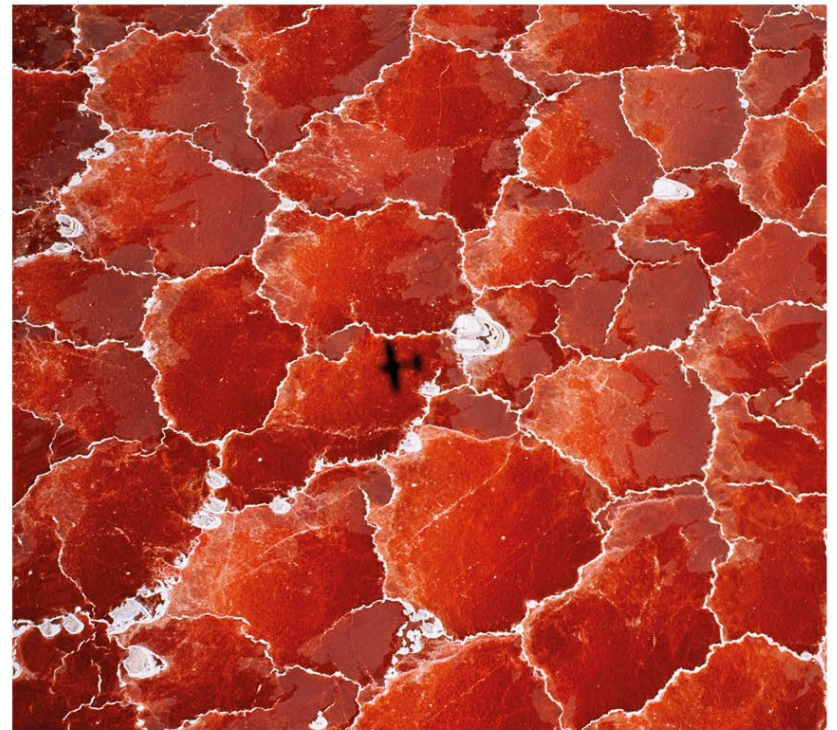
The Black Sea

Location Bulgaria, Romania, Ukraine, Russia, Georgia, and Turkey

This saline water body is described as a trapped ocean basin. It is almost completely surrounded by land but underlain by oceanic crust that was trapped by tectonic plate collisions. To the southwest, it connects to the Mediterranean Sea through the Bosphorus Strait, Sea of Marmora, and Dardanelles Strait. To the north, it connects to the Sea of Azov. Excluding the latter, the Black Sea has an area of 168,300 square miles (436,000 square km) and a maximum depth of 7,218 ft (2,200 m). Many rivers discharge into it, including the Danube and Dnieper.

Other rivers and lakes in Europe

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▲ ACCUMULATION OF CYANOBACTERIA ON LAKE NATRON

Africa

Lake Chad

Location The Chad Basin, Nigeria, Niger, Chad, and Cameroon

This freshwater lake is situated in the Chad Basin, the second largest endorheic (closed) drainage basin on Earth. Between 1963 and 1998, it shrank from 10,000 square miles (26,000 square km) to 520 square miles (1,350 square km), although a small increase followed. Mostly fringed by marshes, it averages just 5 ft (1.5 m) in depth, although this varies from year to year. Lake Chad is fed by rivers such as the Ngadda and Komadugu Yobe and has no outlet. The lake is a vital source of water and fish for local communities.

Lake Nakuru

Location The eastern arm of the Great Rift Valley, western Kenya

One of the ultra-alkaline soda lakes in the bottom of the down-faulted East African Rift Valley, this water body is famous for

supporting hundreds of thousands of flamingos that come to feed on the lake's algae. Nakuru varies in size from 2 to 17 square miles (5 to 45 square km), but its maximum depth is just 10 ft (3 m). About 10,000 years ago, Nakuru and its neighboring lakes (Elmenteita and Naivasha) formed a single deep freshwater lake, which shrank dramatically as the climate became drier, leaving behind three separate remnant lakes.

Lake Natron

Location Arusha region, northern Tanzania

Fed by mineral-rich hot springs and the Ewaso Ng'iro River, Lake Natron is unusual for its deposits of the evaporite minerals natron and trona, its brightly colored waters, and the huge flocks of flamingos that feed and breed there. With water temperatures as high as 140°F (60°C), this shallow soda lake at the bottom of the Gregory Rift (the eastern branch of the East African Rift Valley) is inhospitable for most organisms, but abundant cyanobacteria color its waters deep red or orange.